- A set of standard claims that can be used in the JSON content of the JWT, with defined meanings, such as `exp` to indicate the expiry time and `sub` for the subject, just as you have been using.
- A wide range of algorithms for authentication and encryption, as well as digital signatures and public key encryption that are covered later in this book.

Because JWTs are standardized, they can be used with lots of existing tools, libraries, and services. JWT libraries exists for most programming languages now, and many API frameworks will include built-in support for JWTs, making them an attractive format to use. The OpenID Connect (OIDC) authentication protocol, that's discussed in chapter 7, uses JWTs as a standard format to convey identity claims about users between systems.

The JWT standards zoo

While JWT itself is just one specification (https://tools.ietf.org/html/rfc7519), it builds on a collection of standards collectively known as JSON Object Signing and Encryption (JOSE). JOSE itself consists of several related standards:

- JSON Web Signing (JWS, https://tools.ietf.org/html/rfc7515) defines how JSON objects can be authenticated with HMAC and digital signatures.
- JSON Web Encryption (JWE, https://tools.ietf.org/html/rfc7516) defines how to encrypt JSON objects.
- JSON Web Key (JWK, https://tools.ietf.org/html/rfc7517) describes a standard format for cryptographic keys and related metadata in JSON.
- JSON Web Algorithms (JWA, https://tools.ietf.org/html/rfc7518) then specifies signing and encryption algorithms to be used.

JOSE has been extended over the years by new specifications to add new algorithms and options. It is common to use JWT to refer to the whole collection of specifications, although there are uses of JOSE beyond JWTs.